

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1703

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:30

Annexure A

Constraint Problem Solving

Code of Conduct:

I B.Theerthi (192473926) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

B.Theerthi

Signature of the student:

Annexure A

Constraint Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night)

such that:

Constraints:

i. D1 cannot work Night shift

ii. D2 must work before D3 (Morning < Afternoon < Night)

iii. Only one doctor per shift

iv. D3 cannot work Morning

v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

Step 1: Initial State

No doctors are assigned.

D1 = _

D2 = _

D3 = _

Step 2: Assign D1

Try assigning **Morning** to D1.

D1 = Morning

Constraint Check

- D1 is not assigned Night.
- Shift is available.

Assignment is valid.

Current Assignment:

D1 = Morning

D2 = _

D3 = _

Step 3: Assign D2

Try Morning

D2 = Morning

Constraint Check:

- Morning is already assigned to D1. ✗

This assignment is invalid.

Backtrack and try another shift.

Try Afternoon

D2 = Afternoon

Constraint Check:

- Afternoon is available. ✓

Assignment is valid.

Current Assignment:

D1 = Morning

D2 = Afternoon

D3 = _

Step 4: Assign D3

Try Afternoon

D3 = Afternoon

Constraint Check:

- Afternoon is already assigned to D2. ✗

Invalid assignment.

Backtrack and try another shift.

Try Night

D3 = Night

Constraint Check:

- D3 is not assigned Morning. ✓
- Night is available. ✓
- D2 works before D3.
 - D2 = Afternoon
 - D3 = Night
 - Afternoon comes before Night. ✓

Assignment is valid.

Current Assignment:

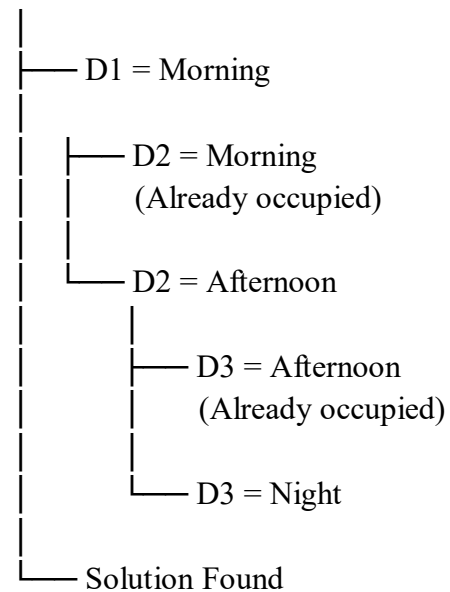
D1 = Morning

D2 = Afternoon

D3 = Night

Backtracking Search Tree

Start



Final Constraint Verification

- D1 is not assigned Night.
- D2 works before D3.
- Only one doctor is assigned to each shift.
- D3 is not assigned Morning.
- Every doctor is assigned exactly one shift.

Final Valid Schedule

Morning → D1

Afternoon → D2

Night → D3

Conclusion

Using the **Backtracking Search algorithm**, invalid assignments were rejected through backtracking, and a valid assignment was found by satisfying all the given constraints.

Final Answer:

- D1 → Morning
- D2 → Afternoon
- D3 → Night

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- Movement allowed: Up, Down, Left, Right**
- Cost of each move = 1**
- Cannot pass through obstacles**
- Use Manhattan Distance heuristic**

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

Given_Constraints

- Move only Up, Down, Left, Right
- Cost of each move = 1
- Cannot pass through obstacles (X)
- Use Manhattan Distance heuristic

Grid 5*5

```
S . . X .  
. X . X .  
. X . . .  
. . X X .  
X . . . G
```

BFS Search Steps

Step 1: Start at **S (0,0)**

Queue: (0,0)

Step 2: Expand (0,0)

Add (1,0) and (0,1)

Queue: (1,0), (0,1)

Step 3: Expand (1,0)

Add (2,0)

Queue: (0,1), (2,0)

Step 4: Expand (0,1)

Add (0,2)

Queue: (2,0), (0,2)

Step 5: Continue expanding nodes until the goal is reached.

Optimal Path

S

↓

(1,0)

↓

(2,0)

↓

(3,0)

→

(3,1)

↓

(4,1)

→

(4,2)

→

(4,3)

→

G

Manhattan Distance

Formula

$$h(n) = |x_g - x| + |y_g - y|$$

At Start (0,0):

$$h = |4 - 0| + |4 - 0| = 8$$

At Goal (4,4):

$$H = 0$$

Total Cost

- Number of moves = **8**
- Cost per move = **1**

Total Cost = 8

Conclusion

Using **BFS**, the robot explores the grid level by level, avoids obstacles, and finds the **shortest path** from **Start (S)** to **Goal (G)** with a **minimum cost of 8**.

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and Rescue survivors. The robot operates in a partially observable environment where some Paths are blocked, and it must make decisions with limited information. The building is represented as a grid of rooms. The robot starts at position S and must reach a Survivor located at G.

Tasks:

A) Analyze all the given constraints and explain how they affect the robot's decision-making Process.

B) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.

C) Demonstrate how the robot applies AI concepts such as:

- ☐ **Intelligent agents**
- ☐ **Rationality**
- ☐ **Environment type**

D) Justify why your chosen approach is the most suitable for solving this problem under the

Given constraints.

(a) The robot must reach the survivor (**G**) safely while following these constraints:

- Move only **Up, Down, Left, Right**.
- Cannot move through blocked cells (obstacles).
- Can observe only adjacent cells (limited sensing).
- Normal movement cost = **1**.
- Risk zone cost = **1 + 2 = 3**.
- Must choose the **minimum-cost and safest path**.
- Updates its map whenever new obstacles are found (online search).

(b) Solution Approach (Uniform Cost Search + Online Search)

Step 1: Start from **S**.

Step 2: Observe adjacent cells.

Step 3: Calculate movement cost.

- Normal cell = 1
- Risk zone = 3

Step 4: Choose the path with the **lowest total cost**.

Step 5: If a new obstacle is detected, update the map and search again.

Step 6: Continue until the robot reaches **G**.

(c) AI Concepts Applied

Intelligent Agent The robot senses the environment, makes decisions, and moves automatically toward the goal.

Rationality The robot always selects the safest and lowest-cost path.

Environment Type

- **Partially Observable** – Only nearby cells are visible.
- **Dynamic** – The environment may change.
- **Sequential** – Each move affects the next move.
- **Discrete** – The environment is represented as grid cells.

(d) Justification

Uniform Cost Search (UCS) with an **Online Search Agent** is the best choice because:

- It finds the **minimum-cost path**.
- It avoids risky and blocked areas.
- It can adapt when new obstacles are discovered.
- It ensures safe and efficient navigation during rescue operations.

Conclusion

The rescue robot uses **Uniform Cost Search** and **online search** to safely reach the survivor while minimizing total cost. This approach satisfies all the given constraints and is suitable for a disaster-affected environment.

RUBRICS FOR CALCULATION:

Criteria	Weight age	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understand s problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear proces s
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and	Understand s problem with minor gaps	Partial understanding; misses key elements	Misinterpret s or unable to understand problem

		requirements accurately			
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